

Run 3 ZH and ZZ Four-Lepton Pairing Study

Associated-Z recovery, two-boson partition fidelity, and region stability

ZH $\rightarrow 4\ell + \text{MET}$ PairingStudy configuration

Final all-file, all-event campaign — 11 August 2026

Abstract

This report compares six reconstructed four-lepton Z/X pairing conventions in two deliberately separate truth domains: the label-sensitive associated-Z pair in $Z(\ell\ell)H(WW \rightarrow 2\ell 2\nu)$ and the label-invariant generator-record two-boson partition in direct-four-lepton ZZ events. The study processes all 780 configured Run 3 input files across five eras in 86 successful jobs, using one pairing-independent quartet, one exhaustive candidate cache, and one event graph. The established physical-mass nearest- m_Z method reaches a raw associated-Z efficiency of 94.1712% and a ZZ partition fidelity of 97.4749%. Source-associated FSR recovery gives only a small positive net gain (+851 ZH and +3 ZZ assignments), while lowering both signal-region and ZZ-control-region acceptance. Resolution-pull variants lose net correct assignments in both domains. The evidence supports retaining the current nearest- m_Z rule.

Primary conclusion. Keep the current physical-mass nearest- m_Z pairing. Core and historical massless rules are effectively identical to it in production (1 changed ZH assignment in 668,539 physical-base events; none in ZZ). Explicit X ranking is exactly redundant after fixing the quartet and Z: the measured XSF $\leftrightarrow$ XDF off-diagonal count is zero.

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1 Physics questions and truth definitions

The same reconstructed pairing rule is studied against two different truth objects. Their denominators and efficiencies are never combined.

1.1 ZH associated-Z truth

The signal is

$$pp \rightarrow ZH, \quad Z \rightarrow \ell^+ \ell^-, \quad H \rightarrow WW \rightarrow \ell \nu \ell \nu.$$

The truth target is the unique hard-process Z lineage that is not a Higgs descendant. Direct prompt $Z \rightarrow ee$ or $Z \rightarrow \mu\mu$ daughters are followed through same-PDG copy chains and must match two distinct quartet leptons one-to-one. A reconstructed candidate is correct only if its unordered selected-Z index pair equals this associated-Z pair. This definition is *label sensitive*: exchanging the associated-Z pair with an SF HWW pair is wrong even when the complete four-lepton partition is unchanged. The complement is independently checked against the two matched HWW leptons.

The $3e1\mu$ and $1e3\mu$ topologies are essential signal channels: the associated Z is SF and the HWW complement is DF. They are retained throughout.

1.2 ZZ two-boson truth

For inclusive ZZ, the direct-four-lepton truth subset requires two distinct hard Z/γ^* neutral-current lineages, each with direct prompt ee or $\mu\mu$ daughters, and a one-to-one match of all four daughters to the quartet. FSR/conversion photons and nonleptonic intermediate decays are not promoted to direct lineages. Correctness is

$$\{\{Z_1, Z_2\}, \{X_1, X_2\}\}_{\text{reco}} = \{\{\ell_1, \ell_2\}, \{\ell_3, \ell_4\}\}_{\text{truth}},$$

where both the individual pairs and the set of pairs are unordered. Swapping the two bosons or the reconstructed Z/X labels is accepted. The $2e2\mu$ partition is flavor distinguishable. For $4e$ and 4μ , identical-particle interference prevents a unique observable pairing; the metric is explicitly generator-record partition fidelity. Odd-flavor quartets remain reconstructed diagnostics but do not enter the direct- 4ℓ ZZ truth denominator.

2 Pairing-independent event contract

The principal denominator is defined before any Z/X assignment:

- the four highest- p_T leptons passing the configured Run 3 tight electron or muon working point;
- strict ordered thresholds $p_T > 25, 15, 10, 10$ GeV;
- total selected charge zero;
- a veto on any fifth common **Lepton** with $p_T \geq 10$ GeV.

This defines **PAIRING_OBJECT_BASE**. The physical baseline **PAIRING_PHYS_BASE** additionally requires the minimum invariant mass among all six quartet pairs to exceed 12 GeV. Neither baseline uses a selected Z, a Z window, X flavor, m_X , MET, or analysis-region membership.

Every unordered OS-SF pair is enumerated once as a Z candidate and X is the two-object complement. Four-vectors, nominal and alternative scores, candidate multiplicity, truth state, and region labels are cached once per event and reused by all histograms.

Table 1: Executable pairing algorithms. All methods consume the same quartet and candidate set.

Code	Name	Ranking rule
0	nearest m_Z	Physical electron/muon masses; minimize $ m_{\ell\ell} - 91.1876 $ with strict-first tie behavior.
1	core massless	Literal fixed-quartet massless convention of core <code>getZXLepIdx</code> .
2	historical massless	Same executable selector as code 1, retaining historical Run-2 source provenance.
3	resolution pull	Minimize $ m_{\ell\ell} - m_Z /\sigma_{m_{\ell\ell}}$ with independent fixed-direction lepton uncertainties.
4	FSR nearest m_Z	Nearest- m_Z ranking after validated forward/reverse source-associated FSR recovery.
5	FSR resolution pull	FSR-recovered mass with the documented lepton-only resolution approximation.

3 Pairing methods

The normalized physical/core/historical equivalence is a proof-level reference, not duplicate executable code. The core `getZAZBLepIdx` method was audited but excluded: it is an auxiliary cross-pair construction, not a universal selector, and retains a known legacy indexing defect and a sentinel inconsistency. Truth oracles are diagnostic ceilings only. No deployable score uses m_X , MET, region membership, or ML.

4 Inputs and production

Inventories, production steps, luminosities, component weights, and source normalizations are materialized dynamically from `../ZZ_CR/year_config.json`. Each logical sample currently resolves to one same-named component with component weight one. Source normalization is 1.0119790366858 for qqZH, 0.9651076466221233 for ggZH, and 1 for ZZ.

Table 2: Final input inventory. The numbers in parentheses are files.

Era	ZH components	ZZ component	Jobs
2022	qqZH HWW (53), ggZH HWW (5)	inclusive ZZ (21)	10
2022EE	qqZH HWW (23), ggZH HWW (19)	inclusive ZZ (38)	9
2023	qqZH HWW (72), ggZH HWW (47)	inclusive ZZ (11)	15
2023BPix	qqZH HWW (58), ggZH HWW (33)	inclusive ZZ (5)	11
2024	qqZH $Z \rightarrow 2\ell$, HWW (161), ggZH $Z \rightarrow 2\ell$, HWW (158)	inclusive ZZ (76)	41
Total	780 files, all events, 10 files/job		86

The final campaign, `pairing_dual_full_packaged_v2_20260811`, freshly compiled every era and streamed all event inputs from `root://eoscms.cern.ch`. Table 3 records the exact FNAL submission. All 86 jobs terminated normally with return value zero, and all five ordinary mkShapesRDF histogram merges completed without a reported `hadd` failure.

An earlier campaign (clusters 3798577–3798581) failed before event processing because the FNAL worker container did not mount the submit host’s `/uscms_data/start.sh`; it produced no physics output and was stopped. After a successful packaged three-process smoke test (cluster 3798585), the final campaign used mkShapesRDF’s deterministic runtime package for code transport only. Event data were still read directly over XRootD.

Table 3: Final Condor campaign on `lpcschedd4.fnal.gov`.

Era	Jobs	Cluster range	Pickle timestamp
2022	10	3798586.0–9	26-08-11 15:54:06
2022EE	9	3798587.0–8	26-08-11 15:57:29
2023	15	3798588.0–14	26-08-11 15:58:10
2023BPix	11	3798589.0–10	26-08-11 15:58:42
2024	41	3798590.0–40	26-08-11 15:59:19

5 Truth and source quality

Table 4: Pairing-independent physical-base truth and alignment diagnostics.

Quantity	ZH	ZZ
Total reconstructed events	668,539	9,179
Direct target record	651,340 (97.4274%)	8,873 (96.6663%)
Truth recoverable / partition valid	651,076 (97.3879%)	8,871 (96.6445%)
Provable source alignment	661,530 (98.9516%)	9,040 (98.4857%)
Resolution and FSR score complete	660,390	9,040
HWW complement valid	650,701 (99.9424% of recoverable)	—
Identical-flavor record ambiguity	—	4,657 (52.4969% of valid)

The combined HWWNano schema does not provide a reliable final `Lepton_genPartIdx`. Historical scale-smearing output can rarely permute final lepton component vectors independently. The study therefore requires a deterministic one-to-one match to coherent pre-scale source objects with valid flavor/origin information. If identity cannot be proved, truth, resolution, and FSR are marked unavailable; no geometric guess is made. The live nearest- m_Z baseline remains evaluable on the final production vectors.

Candidate multiplicity exposes the combinatorics. ZH contains 1,146 events with zero valid Z candidates, 490,807 with two, and 176,586 with four. ZZ has 4,395 two-candidate and 4,784 four-candidate events and no zero-candidate event on `PAIRING_PHYS_BASE`.

6 ZH associated-Z performance

The ZH truth denominator contains 651,076 recoverable events. Table 5 reports raw counts and the minimal signed and absolute weights. The signed weight is luminosity/source normalized and includes only `XSWeight`, `puWeight`, and `METFilter_Common`.

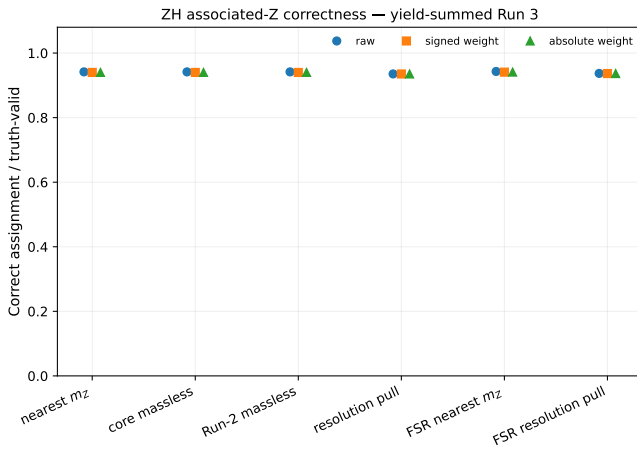
Correct current assignments have a signed binned mean relative $p_T(Z)$ response, $(p_T^{\text{reco}} - p_T^{\text{truth}})/p_T^{\text{truth}}$, of +0.0106; wrong assignments have a much broader negative mean near -0.344 . FSR nearest- m_Z changes these to +0.0103 and -0.340 . The signed binned mean selected m_X changes from 41.64 GeV to 41.41 GeV with FSR nearest- m_Z ; these are reconstructed distributions, not truth- m_X responses.

Table 5: ZH associated-Z correctness on PAIRING_PHYS_BASE.

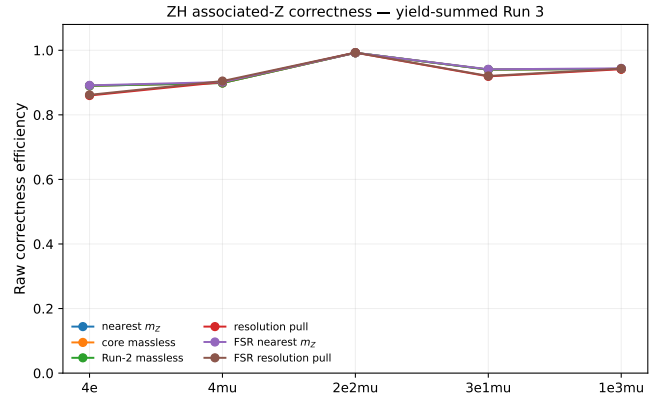
Algorithm	Correct	Raw	Signed	Absolute
Nearest m_Z	613,126	94.1712%	93.9976%	94.0079%
Core massless	613,125	94.1710%	93.9976%	94.0079%
Historical massless	613,125	94.1710%	93.9976%	94.0079%
Resolution pull	609,024	93.5412%	93.5228%	93.5267%
FSR nearest m_Z	613,977	94.3019%	94.0965%	94.1089%
FSR resolution pull	610,086	93.7043%	93.6531%	93.6559%

Table 6: Current nearest- m_Z ZH efficiency by era and topology.

Era	Correct/valid	Raw	Topology	Correct/valid	Raw
2022	7,911/8,399	94.1898%	$4e$	45,622/51,261	88.9994%
2022EE	26,917/28,591	94.1450%	4μ	109,346/121,664	89.8754%
2023	17,770/18,828	94.3807%	$2e2\mu$	156,518/157,749	99.2196%
2023BPix	8,812/9,355	94.1956%	$3e1\mu$	124,380/132,297	94.0157%
2024	551,716/585,903	94.1651%	$1e3\mu$	177,260/188,105	94.2346%



(a) Algorithm comparison.



(b) All five ZH topologies.

Figure 1: Associated-Z correctness in the yield-summed Run 3 signal sample.

7 ZZ partition performance

The ZZ denominator contains 8,871 valid direct-four-lepton partitions.

Table 7: ZZ generator-record partition fidelity on PAIRING_PHYS_BASE.

Algorithm	Correct	Raw	Signed/absolute
Nearest m_Z	8,647	97.4749%	97.7491%
Core massless	8,647	97.4749%	97.7491%
Historical massless	8,647	97.4749%	97.7491%
Resolution pull	8,640	97.3960%	97.6135%
FSR nearest m_Z	8,650	97.5087%	97.7715%
FSR resolution pull	8,639	97.3847%	97.6137%

Current raw fidelity is 97.3171%, 97.0472%, 97.0390%, 98.4529%, and 97.8274% in 2022, 2022EE, 2023, 2023BPix, and 2024. Its topology dependence separates physical interpretations:

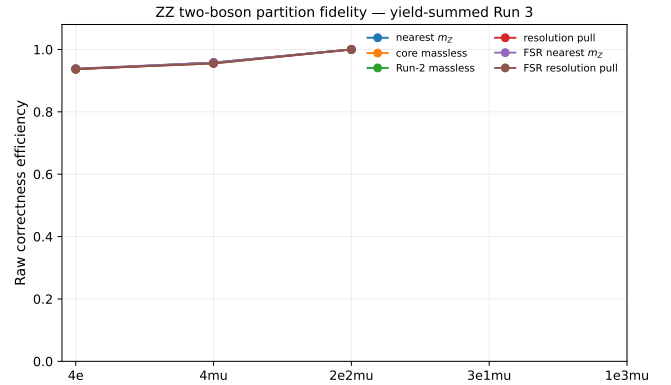
Table 8: Current ZZ fidelity by topology.

Topology	Correct/valid	Raw	Interpretation
$4e$	1,202/1,283	93.6867%	Identical-flavor generator-record convention
4μ	3,229/3,372	95.7592%	Identical-flavor generator-record convention
$2e2\mu$	4,216/4,216	100.0000%	Flavor-distinguishable partition
$3e1\mu$	0/0	—	Not a direct- 4ℓ ZZ truth topology
$1e3\mu$	0/0	—	Not a direct- 4ℓ ZZ truth topology

The current signed binned mean selected m_X is 88.22 GeV; all alternatives remain within 0.11 GeV. A deterministic truth dilepton nearest m_Z is used only for the ZZ $p_T(Z)$ response. This reference label never enters partition-correctness decisions.



(a) Algorithm comparison.



(b) Direct- 4ℓ topology dependence.

Figure 2: ZZ two-boson generator-record partition fidelity.

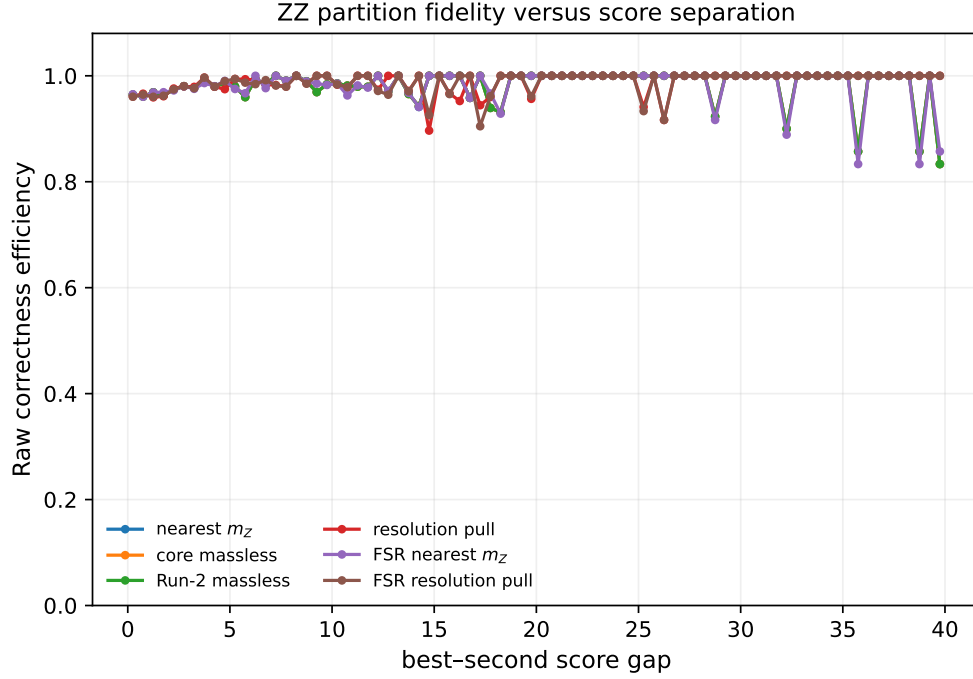


Figure 3: ZZ partition correctness versus the nearest- m_Z score gap. Small gaps identify the more ambiguous candidate configurations.

8 Exact gains, losses, and candidate agreement

Table 9 uses joint event-level states relative to current nearest- m_Z ; it does not infer gains and losses by subtracting marginal efficiencies.

Table 9: Exact raw correctness transitions and candidate agreement.

Domain	Alternative	Wrong→correct	Correct→wrong	Net	Agreement
ZH	Core/historical massless	0	1	−1	99.99985%
ZH	Resolution pull	4,635	8,737	−4,102	96.6343%
ZH	FSR nearest m_Z	1,541	690	+851	98.6083%
ZH	FSR resolution pull	5,877	8,917	−3,040	96.4235%
ZZ	Core/historical massless	0	0	0	100.0000%
ZZ	Resolution pull	20	27	−7	88.6698%
ZZ	FSR nearest m_Z	8	5	+3	96.9169%
ZZ	FSR resolution pull	25	33	−8	87.6239%

FSR nearest- m_Z gives a signed net gain of +0.01250 in ZH and +0.4171 in ZZ. Signed efficiency increases are about 0.099 and 0.022 percentage points. Negative ZH weights are retained explicitly; no negative event weight appears in the final selected ZZ sample, so signed and absolute ZZ efficiencies coincide.



Figure 4: Exact joint raw gain and loss fractions for the resolution- and FSR-aware comparators.

9 Region migration and acceptance

Region definitions reproduce the numerical study selections after the common pairing-independent denominator. Region kinematics always use nominal lepton-only four-vectors; FSR is used only to rank candidates. Thus the migration matrices isolate the pairing decision.

Table 10: Raw analysis-region acceptance on PAIRING_PHYS_BASE.

Method	ZH XSF+XDF SR	ZH ZZCR	ZZ ZZCR
Current/core/historical	492,952 (73.7357%)	1,871 (0.2799%)	6,031 (65.7043%)
Resolution pull	488,146 (73.0168%)	1,861 (0.2784%)	5,977 (65.1160%)
FSR nearest m_Z	490,048 (73.3013%)	1,829 (0.2736%)	5,976 (65.1051%)
FSR resolution pull	488,281 (73.0370%)	1,847 (0.2763%)	5,977 (65.1160%)

For ZH, FSR nearest- m_Z moves 9, 260, and 301 outside events into ZZCR, XSF SR, and XDF SR, while moving 51, 1,735, and 1,730 events in the reverse direction. Resolution pull has 62/872/976 outside-to-region migrations and 72/2,972/3,682 reverse migrations. The combined rule has 68/1,094/1,198 gains and 92/3,102/3,861 losses.

For ZZ, FSR nearest- m_Z moves 4 events from outside to ZZCR but 59 from ZZCR to outside, plus 2 XSF events outside. Resolution pull moves 15 outside events to ZZCR and 69 back; FSR-resolution moves 18 in and 72 out. No alternative produces a direct ZZCR-to-SR or XSF-to-XDF transition. Nontrivial region changes pass through *outside*; X flavor remains invariant.

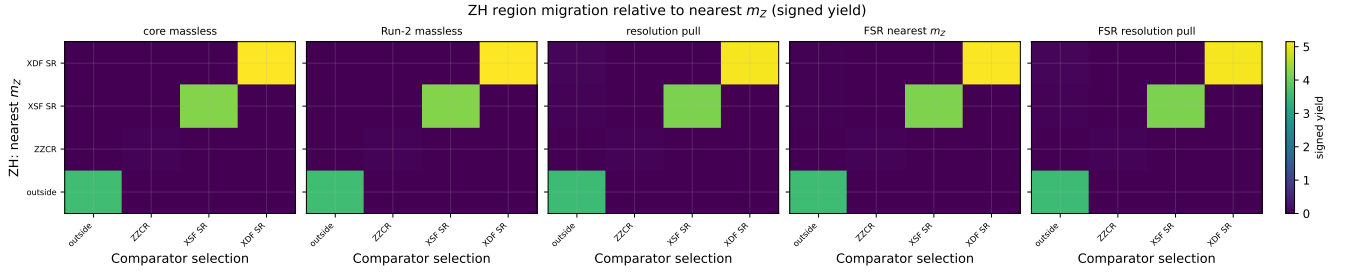


Figure 5: ZH region assignment matrices relative to current nearest- m_Z , using signed yields.

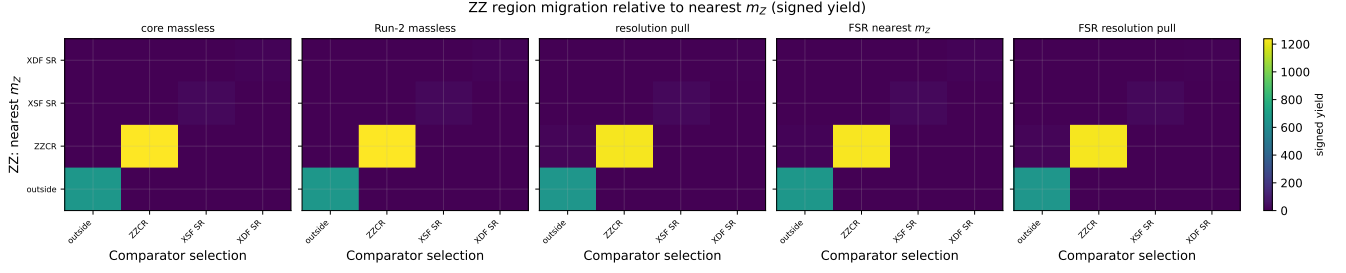


Figure 6: ZZ region assignment matrices relative to current nearest- m_Z , using signed yields.

10 X-ranking theorem and measurement

Let the fixed quartet have total charge zero and let the selected Z be an OS pair. The quartet therefore contains two positive and two negative leptons. Removing one positive and one negative lepton leaves exactly one positive and one negative lepton, so the complement X is necessarily OS. Because only two objects remain after fixing Z, no independent X candidate can be ranked.

The event-level measurement is exact:

- ZH: 667,393/667,393 valid-Z events have the live-style X equal to the two-object complement;
- ZZ: 9,179/9,179 valid-Z events have exact identity;
- the remaining 1,146 ZH events have no valid nearest- m_Z candidate and are not X disagreements; and
- measured physical XSF $\leftrightarrow$ XDF migrations are exactly zero.

Explicit X ranking is therefore redundant within this fixed-quartet study. This result is diagnostic and does not by itself authorize a change to the production ZZ control-region configuration.

11 Validation and limitations

The final implementation passed 37 focused PyROOT/Python tests covering all five topologies, candidate multiplicities, strict tie behavior, the literal massless edge case, source-vector corruption, uncertainty propagation, FSR link validation and deduplication, same-candidate-set availability, ZH and ZZ truth contracts, boson-label-swap invariance, duplicated matches, region migrations, exact gain/loss states, X-flavor closure, and mkShapesRDF's first-logical-axis-fastest histogram unrolling.

Further validation included:

- representative ZH and ZZ schemas for all five eras;
- all five merged ROOT files reopened as non-zombie files with both pairing baselines;
- strict final summary generation with zero warnings;
- yield-summed, rather than percentage-averaged, ALL_RUN3;
- 14 supported plot products in both PDF and PNG, visually inspected;
- Python compilation, shell syntax, and staged whitespace checks.

Truth- m_X response plots were deliberately omitted. There is no unique, defensible truth- m_X target shared by the two domains, and the selected- m_X distribution was not misrepresented as a response. The resolution score neglects correlations and directional uncertainties; the combined FSR-pull score has no photon-resolution term. The study uses minimal nominal MC weights and is not a detector-systematics or analysis-scale-factor study.

12 Recommendation

Retain the current physical-mass nearest- m_Z pairing. It is simple, stable across all five eras and all five ZH topologies, exactly reproduces the live Run 3 convention, and already gives high associated-Z and ZZ-partition fidelity. FSR nearest- m_Z has a real but small net truth gain in both domains; that gain does not compensate for its source/FSR association burden, its fail-closed unavailable population, and its lower signal-region and ZZCR acceptance. Resolution-aware variants are disfavored because they lose net correct assignments in both ZH and ZZ. No production pairing change is justified by these results.

A Reproducibility and artifact locations

The audited starting revision was 1a44b3143560703146039f03a1b6118e9a769243; the completed study was committed as 3659c2e930d58b8a3df387ca9080c9443bb528e8 on ZH_devel.

Relative to this configuration directory, generated products are located at:

- merged ROOT files: `rootFiles/pairing_dual_full_packaged_v2_20260811/<era>/`;
- CSV/JSON reports: `reports/pairing_dual_full_packaged_v2_20260811/`;
- figures: `plots/pairing_dual_full_packaged_v2_20260811/`;
- submission controls and receipts: `condor/pairing_dual_full_packaged_v2_20260811/`;
- submission and merge logs: `submit_pairing_dual_full_packaged_v2_20260811.log` and `merge_pairing_dual_full_packaged_v2_20260811.log`.

Exact inventories and source paths are in `reports/pairing_dual_full_packaged_v2_20260811/sample_inventory.json`. The complete machine-readable result includes ZH and ZZ efficiencies, algorithm agreement, migration matrices, truth-matching diagnostics, X-ranking closure, plot data, and manifests.